AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2012

General Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL TEN Questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a.

i. Simplify $\frac{l-m}{l-n} \times \frac{m-n}{m-l} \div \frac{m-n}{l-n}$. (3 Marks)

ii. Simplify $\frac{1-\frac{1}{a^2}}{1-\frac{1}{a}}$. (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

b. Show that $\frac{7}{\sqrt{3}+\sqrt{2}}$ can be written in the form of $a\sqrt{3}+b\sqrt{2}$.

Hence, state the values of a and b. (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 5 Marks)

a. Factorize $x^2(x+2)+(2x+4)+3x(x+2)$ completely.

b. If $(x - k)$ is exactly divisible by $(2x^3 - 3kx^2 + 3x + k)$, then show that $4k - k^3 = 0$. Hence, find the possible values of k.

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 4 Marks)

a. Simplify $\dfrac{(p+q)}{(p-q)} \div \dfrac{p^2+2pq+q^2}{p^2-q^2}$.

b. Show that the square root of $n^2(n-1)^2+(n-1)^2(2n+1)$ is (n^2-1).

Q.4. (Total 5 Marks)

If $\frac{x}{3} - \frac{2-x}{4} = 1 - \frac{x}{6}$, then find the solution set.

(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 5 Marks)

a. Solve $\frac{1}{m-6} + \frac{1}{m+6} = \frac{m^2}{m^2 - 36}$.

b. If $2x^2 - 8x + 10 = 2(x+m)^2 + n$, then find the value of m and n.

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

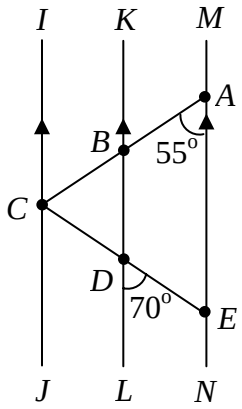
Q.6. (Total 5 Marks)

a. Given that $A = \begin{bmatrix} 3 & -5 \\ 1 & x \end{bmatrix}$; $B = \begin{bmatrix} 2 & -5 \\ y & -3 \end{bmatrix}$ and matrix B is the inverse of A, find the value of x and y.

b. If the determinant of $\begin{bmatrix} 3 & 3 \\ 2 & k \end{bmatrix} + 2\begin{bmatrix} -3 & -2 \\ -4 & -3 \end{bmatrix}$ is k, then find the value of k.

Q.7. (Total 5 Marks)

In the given diagram, if $\angle LDE = 70°$ and $\angle CAE = 55°$, then find

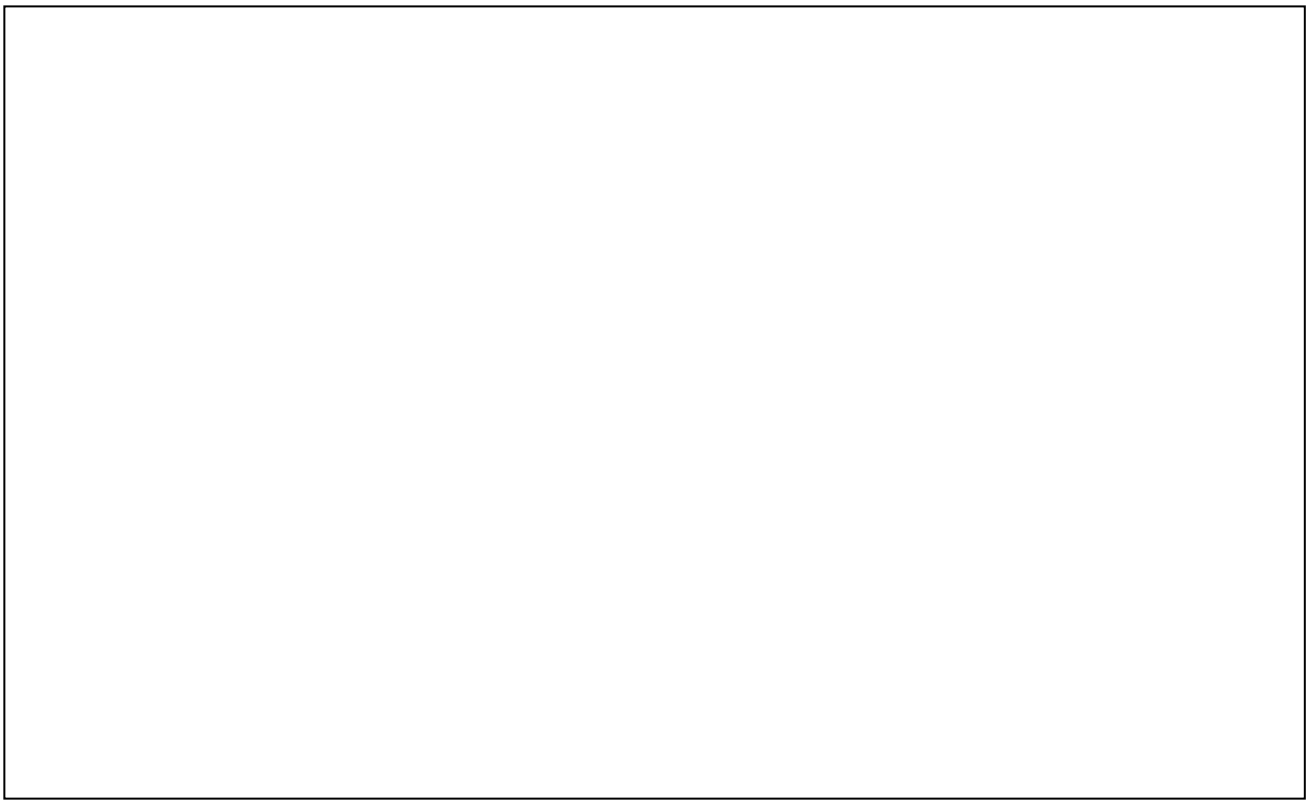


i. $m\angle CAM$. (1 Mark)

ii. $m\angle BCJ$. (4 Marks)

Q.8. (Total 4 Marks)

Draw a tangent line at any point *A* on the circle of radius 2.5 cm with centre *O*.

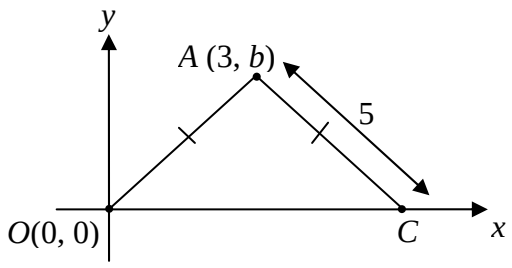


Q.9. (Total 4 Marks)

If the sides of a right angled triangle are k , 3 and $k+1$, then find the value of k, given that $(k + 1)$ is the hypotenuse.

Q.10. (Total 3 Marks)

Given that AOC is an isosceles triangle as shown in the diagram, find the value of b.



END OF PAPER

Please use this page for rough work

Please use this page for rough work